

**WHY THIS CASSETTE****Zyramic Palisade flat sheet membrane with targeted scouring (patent pending)****Selling points and confidence builders**

A one-page reminder for the meeting. Module: Zyramic Palisade flat sheet membrane. Feature: targeted scouring. Full: Zyramic Palisade flat sheet membrane with targeted scouring (patent pending). Public word is scour / scour manifold — never aeration, diffuser, or bubble.

The product

- **Targeted scouring, every plate**
Detachable modular scour manifold on the bottom of the plate. Check valves keep sludge and mixed liquor out of the air path. Secure in service, easy to change.
- **Dedicated scour, 24 / 7**
Air that stays on, every plate, to protect the cassette. Not plant blowers cycling on and off all day.
- **A real support plate**
Polypropylene plate holds the PES sheet so edges do not fray or delaminate the way a thin-film sheet can. Lightened with a blowing agent. American-made PP.
- **PES that is already in the field**
Thermo Fisher Scientific sheet. Years in the market. An advanced product. A major German producer already runs it.
- **Sized as a commercial module**
130 plates single deck, 260 plates double deck. About 1.05 m² per plate (both faces). Panel height 41.2 in so the top stays clean. Scour intensity 10 L/min per m² of membrane.
- **Made here, to BABA**
Cut, weld, assemble, and pack in Poughkeepsie. BABA at the module. Do not claim BABA at system level.

Why it runs cleaner

- **Uniform velocity between plates**
Air at the root of every channel — not a lucky subset. Plates are not packed too close to scour, and not too tall.
- **Less clogging, slower TMP rise**
The operating result of per-channel scour. Less time in the tank. Do not invent a kWh, a percent energy, or a plant TMP curve.
- **The problems the field already knows**
Early clogging (sludging and ragging), hair in hollow fiber, thin-film delamination, broken fibers, expensive ceramics. Seasoned plants often get only 3 to 5 years. The current state of the art is not finished. We have to do better.

Confidence builders

- **Independent CFD**
Targeted scouring versus a conventional bottom plenum. Same plates. Channel-base liquid shear about 4× on the targeted-scouring path. Hydrodynamics, not plant data. Credit as independent if credited at all.
- **Independent FEA**
Module in operation and in lift-out. Worst case included: wetted, double-deck.
- **People who have done this**
Filtration people and structural engineers on staff, with MBR experience. Founders: more than thirty years on plants in the United States and abroad.
- **Qualification, not a chase**
Customers have to want a plate. Do not compete with thin-film on \$/m² or footprint. Story is Palisade with targeted scouring, BABA, no PFAS, a real plate.

KEEP THESE STRAIGHT · sales talk vs public page

Public word: modules, not systems. Scour / scour manifold — never aeration, diffuser, or bubble. Do not name Kubota, MANN+HUMMEL, BIO-CEL, Rubicon, Semicorex, Tangent, or any OEM on the public site. Do not invent a Zyramic service-life number. 3–5 years is the incumbent observation. Do not say never clogs. Do not claim aeration built into the plate. Do not offer the CFD report for download. Ceramic origin stays on the quotation. Public address is 10 Tucker Dr — never 36 Ridgeline Drive.

